

Where XFOIL Stops Being Trustworthy

A validation of XFOIL against UIUC wind tunnel measurements at the Reynolds numbers a Formula SAE wing actually operates at

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Abstract

XFOIL is the tool nearly every Formula SAE team uses to choose an airfoil, and its published validations are at Reynolds numbers five to twenty times higher than an FSAE wing ever sees. This study compares XFOIL against the UIUC low-speed wind tunnel measurements of Selig and McGranahan on three airfoils — E387, FX 63-137 and SD2030 — from $Re = 100,000$ to $500,000$, in two configurations each: clean, and with a zigzag boundary-layer trip near the leading edge.

With transition forced at the measured trip stations, XFOIL has no free parameter, and it predicts drag to within 2% at $Re = 350,000$ and to within 5–23% at $Re = 100,000$. With transition free, the low-Reynolds-number answer is not a property of the airfoil at all: sweeping the amplification factor N_{crit} from 7 to 13 — a range entirely permitted by the tunnel's stated turbulence intensity of “less than 0.1%” — moves the predicted effect of tripping by roughly 50 percentage points at $Re = 100,000$ and reverses its sign on all three airfoils. Every airfoil crosses the measured value somewhere inside that permitted range.

Repeating that sweep at every Reynolds number bounds where the problem lives. The range of answers N_{crit} can produce collapses from 23–67 percentage points at $Re = 100,000$ to 3–6 points at $Re = 500,000$. Below $Re \approx 200,000$ the measurement is reachable by choosing N_{crit} , so agreement there is not evidence; above $Re \approx 350,000$ the choice barely matters and the default is harmless.

Decomposing the prediction by surface identifies the mechanism. Essentially the entire predicted trip penalty is manufactured on the lower surface, where XFOIL at $N_{crit} = 9$ predicts laminar flow all the way to the trailing edge; tripping the upper surface moves transition forward by more than half a chord and changes drag by less than 1%. Raising N_{crit} enlarges the upper-surface separation bubble until tripping it becomes a net saving — which is what the wind tunnel measures. XFOIL therefore has the mechanism right and the bubble magnitude wrong at its default N_{crit} , which is a narrower and more useful criticism than an unexplained discrepancy.

Two method results are reported alongside, because both changed answers during the study: the choice of angle-of-attack averaging window moves results by more than the measurement uncertainty does, and three of 182 XFOIL solutions failed to converge while still reporting a drag, one of which by itself reversed the sign of a headline number.

1. What question this answers

An FSAE wing element of 250 mm chord at 15 m/s runs at a chord Reynolds number near 250,000. Published airfoil data — the NACA tables, most textbook polars — sits at $Re = 3,000,000$ to $6,000,000$. At those Reynolds numbers the boundary layer transitions to turbulence near the leading edge and stays attached. At 250,000 it often does not: the laminar boundary layer separates, transitions in the free shear layer, and reattaches, forming a laminar separation bubble whose drag can exceed the skin friction of the whole rest of the airfoil. Whether a design tool captures this is not a detail; it is the difference between a wing that works and one that does not.

So the question is not “is XFOIL accurate?” but “over what range of Reynolds number is XFOIL accurate enough to design with, and what does it get wrong outside that range?”

Testing that with drag alone is weak, because an error in predicted drag can come from the transition model, the panel method, the geometry, or the tunnel corrections, and they cannot be separated. Tripping the boundary layer gives a cleaner test. A trip forces transition at a fixed, known chordwise station. Comparing the same airfoil in the same tunnel with and without one isolates a single physical question — what does forcing transition do to the drag — and most of the systematic errors on both sides cancel, because they are common to both configurations.

It has a second virtue that only became clear late in the study, and that now carries most of the argument: forcing transition removes XFOIL's only significant free parameter. The tripped configuration is the one comparison in this study that cannot be tuned.

2. The experimental data

2.1 Facility and models

All measurements are from the UIUC low-speed subsonic wind tunnel as reported in Selig & McGranahan (2004), the fourth volume of the Summary of Low-Speed Airfoil Data series. The test section is nominally 2.81 ft high by 4.0 ft wide. Models are 1 ft chord, built by Y. Tinel. Lift is measured with a force balance; drag is obtained from the momentum deficit in a wake rake, and the reported drag coefficient at each angle of attack is the average of four spanwise wake profiles.

2.2 The two configurations

Clean: the model as built, with free transition.

Tripped: three-dimensional zigzag trips affixed near the leading edge, upper surface at 2% chord and lower surface at 5% chord, trip height-to-chord ratio $h/c = 0.11\%$. Selig designates this geometry “zigzag trip type F.” The chordwise locations are stated in the header of every tripped data file used here, so they are taken from primary source rather than inferred.

2.3 Stated measurement uncertainty

Section 2.4 of Volume 1 gives the uncertainty analysis, following Coleman & Steele. For the drag coefficient the contributions are: data-acquisition instruments below 1%, measurement repeatability below 1.5%, and spanwise variation, estimated from a 95% confidence interval on the E374 spanwise drag results at $\alpha = 4^\circ$, of 3% at $Re = 100,000$ falling to approximately 1.5% at and above $Re = 200,000$. Lift coefficient uncertainty is 1.5%; angle of attack, 0.08° .

This matters more than it looks. An earlier draft of this study used the spread across the four spanwise wake stations as the experimental uncertainty, giving roughly 20% at $Re = 100,000$. That is a range statistic describing real spanwise flow non-uniformity, not the uncertainty in the mean of four samples, and it is about seven times too wide. Every significance claim here uses Selig's figures.

2.4 Stated flow quality

Section 2.1 of Volume 1, describing the tunnel's settling chamber, honeycomb and four anti-turbulence screens:

“The turbulence intensity has been measured to be less than 0.1%, which is sufficient for low Reynolds number airfoil measurements.”

That sentence is an upper bound and nothing more, and section 5.4 shows that the difference between the bound and a plausible actual value is large enough to determine the study's headline result.

2.5 Data excluded, and why

- SH3055 — clean and tripped drag both available, but the airfoil coordinates are not in any database reachable here, and they are printed only in an appendix of the NREL report that could not be retrieved. Without coordinates there is no XFOIL side to compare against.
- S834 — tripped drag available, clean drag not. No pair, no comparison.
- S822 — no Volume 4 files obtained. The Volume 1 files for this airfoil are a different test campaign and are not comparable.
- SH3055 tripped, Reynolds block labelled 181,773 — excluded on inspection. Its last two rows ($\alpha = 13.37^\circ$ and 14.36°) are byte-identical to rows in the $Re = 100,002$ block of the same file, and there is an unexplained gap in α from 11.28° to 13.37° . Whatever produced that block, it is not four independent wake traverses at $Re = 181,773$.
- Pitching moment — unusable throughout. The 21-point and 33-point Reynolds blocks in several files show a C_m range of about 0.004 where full sweeps of the same airfoil show 0.05 to 0.2. These blocks went through a different reduction and cannot be mixed with the others.

Three airfoils survive: E387, FX 63-137 and SD2030, at 100,000 / 200,000 / 350,000 / 500,000, plus 150,000 for the FX 63-137.

2.6 How well does this tunnel agree with other tunnels?

The study's largest structural weakness is that every measurement comes from one facility. Volume 1 section 2.5 provides the only available check: drag polars for the E387 measured at NASA Langley LTPT, Delft, Stuttgart and UIUC, plotted together in Figure 2.10 at $Re = 60,000, 100,000, 200,000$ and $300,000$. The values below are read off that scanned figure at $Cl = 0.8$ and are good to roughly ± 0.0005 in Cd , which is the reading precision, not the measurement precision.

At $Re = 100,000$, UIUC, NASA Langley and Delft agree closely — all near $Cd = 0.022$ to 0.023 — while Stuttgart reads about 0.030 , some 30% higher, with a pronounced bulge over $Cl = 0.5$ to 0.9 that is the signature of a larger laminar separation bubble. At $Re = 200,000$ the four span roughly $Cd = 0.011$ to 0.014 , about 25% of the mean; excluding Stuttgart, about 13%. At $Re = 300,000$ only UIUC and NASA Langley are shown and they agree within a few percent.

Two consequences, and they cut in opposite directions. Against the study's low-Reynolds-number findings the check is favourable: XFOIL's E387 clean-drag error at $Re = 100,000$ is -15.8% , several times the 3% or so that separates the three consistent facilities, so that finding is not an artefact of one tunnel. Against the high-Reynolds-number findings it is not favourable at all: at $Re = 200,000$ this study reports XFOIL within 1.3% of UIUC on the E387, while the facilities differ from each other by ten times that. Agreement with one tunnel at the few-percent level is not agreement with reality, and no claim of few-percent accuracy anywhere in this document should be read as one.

Selig attributes part of the scatter to the models rather than the tunnels, and names a problem that bears directly on this study: “the E387 from the UIUC tests was slightly decambered and warped at the trailing

edge, which has an effect on the results.” The XFOIL runs here use nominal database coordinates. For the E387 specifically, the computed and measured airfoils are known to be different shapes.

3. The computational method

3.1 XFOIL configuration

XFOIL 6.99, viscous mode. Iteration limit 300. Graphics disabled (PLOP / G F), without which XFOIL aborts in a headless environment.

The clean case is run with free transition. The tripped case is run with forced transition specified at the measured trip stations:

XTR 0.02 0.05 upper surface 2% chord, lower surface 5% chord

Each angle of attack is run in its own XFOIL subprocess with a 25-second timeout. XFOIL run as a continuous sweep carries state between points, so a single bad solve can poison everything after it; one point per process removes that coupling entirely at the cost of some speed.

3.2 Ncrit, and why it gets its own section

XFOIL locates transition with the e^N method. Disturbances in the laminar boundary layer are amplified as they travel downstream, and transition is declared when the most-amplified disturbance has grown by a factor of e raised to Ncrit — at Ncrit = 9, about 8,100 times. The value depends on how large the disturbances were to begin with, which is to say on the turbulence level of the oncoming flow. Drela's guidance runs from 4–8 for a dirty tunnel through 9 for an average one to 12–14 for a sailplane in free air.

Nine is XFOIL's default. The first revision of this study used it without comment, which was the largest unexamined assumption in the work. Mack's correlation relates the amplification factor to turbulence intensity:

$$N_{crit} = -8.43 - 2.4 \ln(Tu)$$

Applied to the UIUC statement in section 2.4, $Tu < 0.1\%$ gives $N_{crit} > 8.1$ and places no upper bound at all. A turbulence intensity of 0.05% gives 9.8; 0.02% gives 12.0. A tunnel with a honeycomb and four anti-turbulence screens could plausibly be anywhere in that range. The source documentation does not pick a value, and section 5.4 reports what happens when it is treated as the unknown it is.

Baseline results in sections 5.1 to 5.3 use Ncrit = 9, so that they are comparable with the first revision and with the way almost everyone runs XFOIL. Section 5.4 is the sensitivity study.

3.3 Geometry and panel convergence

Coordinates come from the AeroSandbox airfoil database and are repanelled to 120 points per surface (239 total) before use. This is not cosmetic. Database coordinate sets are often coarse — the stock E387 at 61 points would not run in XFOIL at all, and the FX 63-137 at 97 panels gave a $Re = 100,000$ drag error of +16.9% which became –9.5% at 239 panels, reversing the sign of the disagreement with experiment. A panel count too low does not announce itself; it returns a converged, plausible, wrong number.

Panel convergence was therefore verified independently for each airfoil. For SD2030 at $Re = 100,000$:

points per side	total panels	XFOIL clean Cd	trip effect
40	79	0.01965	+11.2%
60	119	0.01964	+11.6%
90	179	0.01948	+12.2%
120	239	0.01945	+12.4%

points per side	total panels	XFOIL clean Cd	trip effect
160	319	0.01944	+12.5%

Table 4. Panel convergence, SD2030, $Re = 100,000$. The last doubling moves the answer by 0.1 percentage points, so 239 panels is converged. Every result in this study uses 239.

3.4 Screening, in two stages

Screen one is the residual. XFOIL prints a result whether or not it converged. A solution that has exhausted its iteration limit while the residual is still large will still report a lift and a drag, and nothing in the returned numbers marks it as invalid. Harvesting the last printed value — the obvious way to script XFOIL, and the way this study did it at first — silently collects those. A solution is accepted only if the final iteration index is below the limit and the final rms residual is below 1×10^{-4} , with no convergence failure reported after it. Rejected points are dropped from both the experimental and the computational average, so the two are always compared over an identical set of angles of attack.

Screen two is physics, and it exists because screen one is not sufficient. Forcing transition with XTR bypasses the amplification model, so tripped drag must be exactly independent of N_{crit} . Any tripped point that moves when N_{crit} moves has converged to a different solution branch — converged, low residual, and wrong. No residual test can detect this; only the invariant can. In the N_{crit} sweep of section 5.4, one point (FX 63-137, $Re = 500,000$, $\alpha = 6^\circ$) returned a tripped drag of 0.00991 at $N_{crit} = 13$ against 0.02135 at every other N_{crit} , a factor of two apart in a quantity that cannot legitimately vary at all. It is rejected on that ground alone.

The tolerance is 10%. Accepted angles show tripped-drag variation of 0–3% across N_{crit} , which is ordinary numerical scatter; the rejected point showed 62%. A tighter tolerance of 0.5% was tried first and discarded because it threw away five sound angles along with the bad one.

Three of 182 points failed: FX 63-137 at $Re = 100,000$ and $\alpha = 8^\circ$ (clean), FX 63-137 at $Re = 500,000$ and $\alpha = 8^\circ$ (tripped), and E387 at $Re = 200,000$ and $\alpha = -2^\circ$ (tripped). The first of these had been returning $C_d = 0.119$ against a measured 0.032 — nearly four times the experiment — with the residual still at 0.23 and the drag oscillating between 0.115 and 0.122 from one iteration to the next. Including it flipped the XFOIL trip effect for that airfoil and Reynolds number from +4.6% to -9.5%, which is to say it reversed the conclusion. Three points out of 182 is a 1.6% failure rate and one wrong headline.

4. The comparison procedure

4.1 Matching

Wind tunnel runs do not land on round Reynolds numbers. Blocks are matched to a nominal Reynolds number when within 6% of it, and the XFOIL run uses the measured value, not the nominal one. Experimental drag is linearly interpolated onto the comparison angles from the measured sweep; no extrapolation beyond the measured range is permitted.

4.2 The metric

For each angle of attack, the trip effect is the fractional change in drag coefficient caused by the trip, and the reported figure is the mean of those per-angle values over the comparison window:

$$\text{trip effect} = \text{mean over } \alpha \text{ of } (C_{d,\text{tripped}}(\alpha) - C_{d,\text{clean}}(\alpha)) / C_{d,\text{clean}}(\alpha)$$

The same expression is evaluated twice, once on the wind tunnel data and once on the XFOIL output, over the same angles. A positive value means the trip added drag.

4.3 Window sensitivity

The window is a free choice, and an early version of this study used $\alpha = 0^\circ$ to 8° without justification. Every result is therefore recomputed over six windows — 0–4, 0–8, 2–8, 0–10, –2–8 and 4–10 degrees — and reported as the full spread across all six. Results narrated in the text use the 0–8 window; results claimed as findings must hold across all six.

This turned out to matter more than the measurement uncertainty. The measured trip effect on the E387 at $Re = 500,000$ ranges from +46.4% to +77.3% depending only on which window is chosen, against a measurement uncertainty near 2%. At high Reynolds number the dominant uncertainty in this study is an analysis choice, not the instrument.

4.4 The overlap criterion

A disagreement is claimed only when the experimental range across the six windows and the XFOIL range across the six windows do not intersect. This is deliberately conservative: it treats the window choice as an unknown rather than a nuisance to be averaged away, and it costs several results that looked significant under a single-window comparison.

5. Results

5.1 Clean configuration

airfoil	Re	n	XFOIL error	range over 6 windows	Selig 95% CI
E387	100,000	5	-15.8%	[-30.2, +0.1]	±3.0%
E387	200,000	5	+1.3%	[-6.5, +6.4]	±1.5%
E387	350,000	5	+7.9%	[-4.9, +11.2]	±1.5%
E387	500,000	5	+10.5%	[-4.6, +15.6]	±1.5%
FX 63-137	100,000	4	-50.0%	[-53.8, -34.8]	±3.0%
FX 63-137	150,000	5	-14.3%	[-17.1, -6.2]	±1.5%
FX 63-137	200,000	5	-7.8%	[-11.0, +0.2]	±1.5%
FX 63-137	350,000	5	-2.3%	[-7.2, +9.3]	±1.5%
FX 63-137	500,000	4	-3.1%	[-5.2, +8.9]	±1.5%
SD2030	100,000	5	+6.2%	[+1.8, +10.1]	±3.0%
SD2030	200,000	5	+1.8%	[-2.0, +3.6]	±1.5%
SD2030	350,000	5	-0.6%	[-2.5, +0.5]	±1.5%
SD2030	500,000	5	-1.6%	[-4.8, -0.1]	±1.5%

Table 1. Clean-configuration drag at $N_{crit} = 9$: mean per-angle percentage error of XFOIL against the wind tunnel. Negative means XFOIL predicts less drag than was measured. n is the number of angles of attack surviving convergence screening.

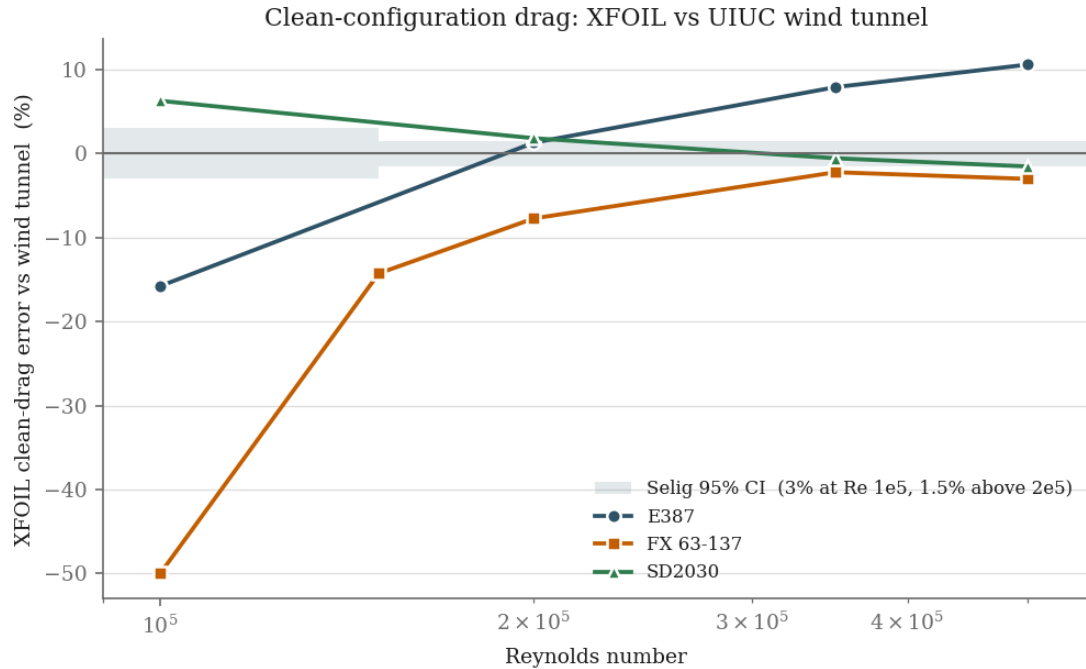


Figure 1. Clean-drag error against Reynolds number, $0-8^\circ$ window, with Selig's stated 95% confidence interval shaded. Values are the mean per-angle errors of Table 1.

XFOIL's clean-drag error is not a simple function of Reynolds number and does not have a consistent sign across airfoils. It under-predicts badly on the FX 63-137 at low Re — by half, at Re = 100,000, where that airfoil carries a large laminar separation bubble. It under-predicts on the E387 at Re = 100,000 and then over-predicts above it. On the SD2030 it is within a few percent everywhere. Section 5.4 shows why this table has no stable message: every number in it moves when Ncrit moves.

5.2 Tripped configuration — the parameter-free comparison

airfoil	Re	n	XFOIL error	range over 6 windows
E387	100,000	5	+4.3%	[−8.3, +16.8]
E387	200,000	5	+19.8%	[+12.7, +27.5]
E387	350,000	5	+1.9%	[+0.8, +5.6]
E387	500,000	5	−2.0%	[−2.9, +0.2]
FX 63-137	100,000	4	−18.5%	[−37.6, +30.3]
FX 63-137	150,000	5	+19.9%	[+15.8, +25.4]
FX 63-137	200,000	5	+19.1%	[+2.7, +27.3]
FX 63-137	350,000	5	−0.8%	[−13.1, +5.1]
FX 63-137	500,000	4	−4.4%	[−21.8, −0.0]
SD2030	100,000	5	+23.0%	[+15.6, +28.2]
SD2030	200,000	5	+16.7%	[+7.2, +23.0]
SD2030	350,000	5	−1.4%	[−2.6, −1.3]
SD2030	500,000	5	−5.0%	[−7.9, −3.7]

Table 2. Tripped-configuration drag: same quantity as Table 1, with transition forced at the measured trip stations. Forcing transition bypasses the e^n model, so unlike Table 1 these numbers do not depend on Ncrit.

This is the table that carries the study. With transition forced there is nothing left to tune: the error is what it is. At Re = 350,000 and above XFOIL is within about 5% on all three airfoils and within 2% on most points. At Re = 100,000 it is off by +4.3%, −18.5% and +23.0% on the three airfoils respectively — large, inconsistent in sign, and not attributable to any parameter choice.

5.3 The effect of the trip, at Ncrit = 9

airfoil	Re	exp	exp range	XFOIL	XFOIL range	★
E387	100,000	−11.4%	[−16.1, −8.6]	+9.9%	[+6.0, +11.9]	★
E387	200,000	+15.4%	[+6.0, +17.2]	+38.0%	[+24.2, +44.0]	★
E387	350,000	+50.8%	[+30.8, +58.5]	+45.7%	[+23.4, +68.2]	
E387	500,000	+66.4%	[+46.4, +77.3]	+52.7%	[+25.3, +81.4]	
FX 63-137	100,000	−25.9%	[−33.7, −20.1]	+4.6%	[−5.2, +34.9]	★
FX 63-137	150,000	+0.3%	[−12.6, +26.0]	+38.6%	[+24.1, +53.6]	
FX 63-137	200,000	+19.0%	[−1.8, +58.3]	+49.6%	[+39.5, +57.7]	
FX 63-137	350,000	+59.0%	[+43.7, +93.6]	+59.9%	[+51.6, +60.6]	
FX 63-137	500,000	+67.5%	[+61.8, +108.0]	+64.6%	[+44.8, +67.6]	

airfoil	Re	exp	exp range	XFOIL	XFOIL range	★
SD2030	100,000	-3.0%	[-9.6, +0.4]	+12.4%	[+6.0, +13.9]	★
SD2030	200,000	+19.0%	[+15.5, +25.4]	+36.6%	[+25.0, +51.3]	
SD2030	350,000	+59.4%	[+33.0, +85.4]	+58.1%	[+30.2, +84.8]	
SD2030	500,000	+73.0%	[+33.6, +106.4]	+66.8%	[+29.2, +98.7]	

Table 3. Effect of the trip on drag, measured and predicted at Ncrit = 9, in the 0–8° window with the full spread over all six windows. A star marks a Reynolds number where the two ranges do not intersect.

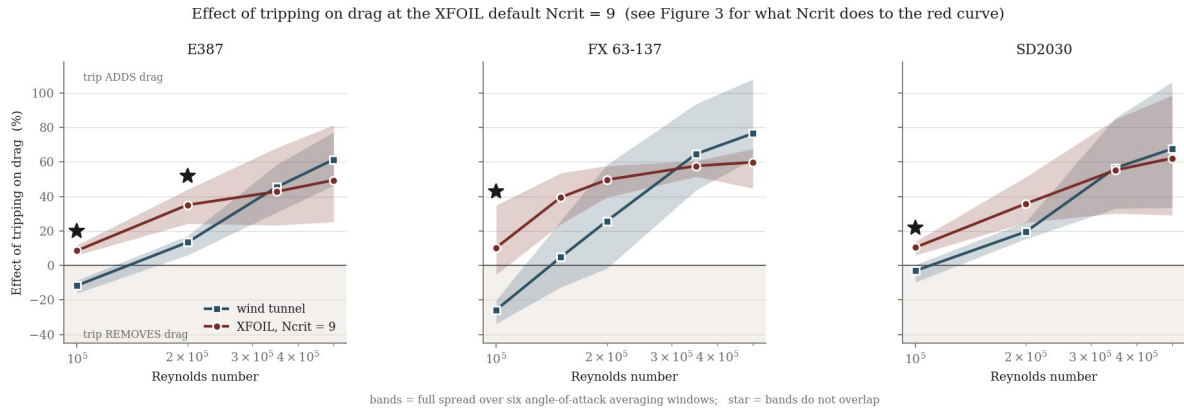


Figure 2. The same comparison plotted. Shaded bands are the full spread over the six averaging windows; lines are their means. Stars mark non-overlapping cases.

At Ncrit = 9 the ranges are disjoint at Re = 100,000 on all three airfoils and overlap on all three by Re = 350,000. Taken alone this looks like a clean statement about XFOIL. Section 5.4 shows that it is a statement about Ncrit.

5.4 Ncrit sensitivity

Before the sweep, a control. XTR forces transition at a fixed station and bypasses the amplification calculation entirely, so tripped drag must be exactly independent of Ncrit. If it is not, the setup is wrong. It is: the E387 tripped drag is 0.02225 and the SD2030 tripped drag 0.02132 at Ncrit = 7, 9, 11 and 13, identical to five significant figures.

airfoil	Ncrit	clean Cd	tripped Cd	XFOIL trip effect	measured
E387	7	0.01857	0.02225	+22.1%	-11.4%
E387	9	0.02041	0.02225	+9.9%	-11.4%
E387	11	0.02388	0.02225	-6.5%	-11.4%
E387	13	0.03296	0.02225	-29.3%	-11.4%
FX 63-137	7	0.02286	0.02982	+29.3%	-25.9%
FX 63-137	9	0.02613	0.02708	+4.6%	-25.9%
FX 63-137	11	0.03527	0.03030	-7.6%	-25.9%
FX 63-137	13	0.04830	0.02945	-29.1%	-25.9%
SD2030	7	0.01819	0.02132	+19.9%	-3.0%

airfoil	Ncrit	clean Cd	tripped Cd	XFOIL trip effect	measured
SD2030	9	0.01945	0.02132	+12.4%	-3.0%
SD2030	11	0.02109	0.02132	+4.2%	-3.0%
SD2030	13	0.02300	0.02132	-3.1%	-3.0%

Table 5. Ncrit sensitivity at $Re = 100,000$, $0-8^\circ$ window. The tripped column is constant by construction; the clean column and therefore the predicted trip effect are not.

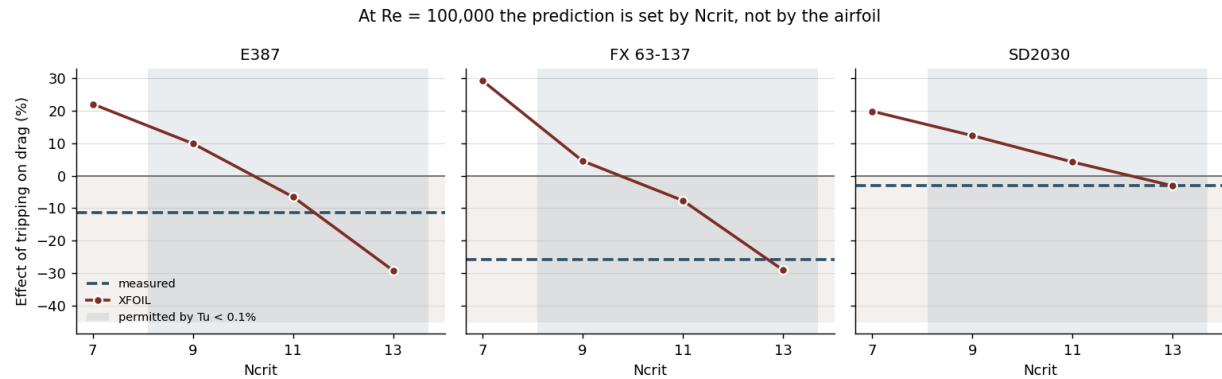


Figure 3. Predicted trip effect against Ncrit, with the measured value as a dashed line and the range permitted by $Tu < 0.1\%$ shaded. All three airfoils cross their measurement inside the permitted range.

The E387 clean drag rises by 77% from Ncrit 7 to 13 while the tripped drag does not move at all, and the predicted trip effect travels from +22.1% to -29.3%, crossing the measured -11.4% on the way. The same happens on the other two. The three curves are close to parallel: changing the airfoil barely moves the picture, and changing Ncrit moves all of it.

Both endpoints of that sweep are admissible. Mack's correlation on the tunnel's stated $Tu < 0.1\%$ permits Ncrit from 8.1 upward with no stated ceiling, so a value that makes XFOIL agree with the measurement and a value that makes it disagree are equally defensible from the source documentation.

The obvious next question is whether this is a low-Reynolds-number problem or a problem everywhere, and revision 2 of this document asserted the former without testing it. Repeating the sweep at every Reynolds number settles it.

airfoil	Re	n	measured	N = 7	N = 9	N = 11	N = 13	spread
E387	100k	5	-11.4%	+22.1%	+9.9%	-6.5%	-29.3%	51.4 ✓
E387	200k	5	+15.4%	+41.0%	+38.0%	+31.3%	+22.0%	19.0
E387	350k	5	+50.8%	+46.7%	+45.7%	+44.5%	+42.2%	4.5
E387	500k	5	+66.4%	+53.0%	+52.7%	+50.9%	+47.9%	5.1
FX 63-137	100k	2	-0.7%	+9.5%	-13.1%	-42.9%	-57.9%	67.3 ✓
FX 63-137	150k	4	+1.2%	+49.9%	+45.0%	+35.9%	+20.3%	29.5
FX 63-137	200k	5	+19.0%	+51.4%	+49.6%	+43.7%	+34.4%	17.0
FX 63-137	350k	5	+59.0%	+55.9%	+59.9%	+60.1%	+58.0%	4.3 ✓
FX 63-137	500k	2	+56.6%	+67.4%	+69.6%	+68.2%	+63.5%	6.1
SD2030	100k	5	-3.0%	+19.9%	+12.4%	+4.2%	-3.1%	23.0 ✓
SD2030	200k	5	+19.0%	+43.4%	+36.6%	+27.3%	+18.3%	25.1 ✓

airfoil	Re	n	measured	N = 7	N = 9	N = 11	N = 13	spread
SD2030	350k	4	+64.9%	+66.0%	+65.5%	+61.0%	+51.9%	14.1 ✓
SD2030	500k	5	+73.0%	+64.2%	+66.8%	+67.0%	+64.9%	2.9

Table 6. Predicted trip effect against Ncrit at every Reynolds number, 0–8° window, both screens applied. Spread is the height of the Ncrit-reachable range in percentage points; a check mark means the measured value falls inside that range, so some admissible Ncrit reproduces it. Rows with n below 3 (FX 63-137 at 100,000 and 500,000) rest on two angles of attack and should not be leaned on.

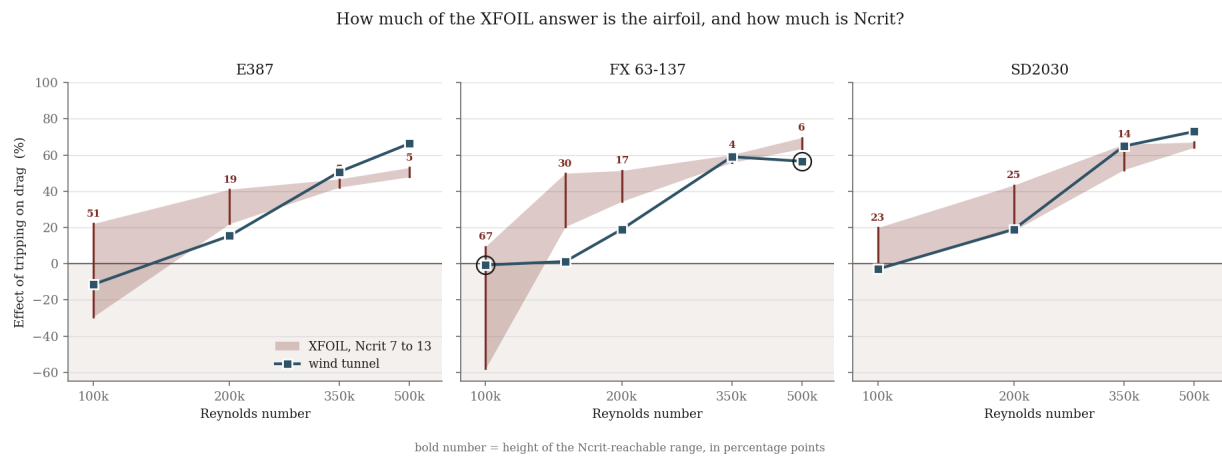


Figure 4. The Ncrit-reachable range against Reynolds number. Rings mark the two weak rows of Table 6.

The range collapses with Reynolds number: 23 to 67 percentage points at $Re = 100,000$, 17 to 30 at $Re = 150,000$ – $200,000$, and 3 to 6 at $Re = 500,000$. Where the range is wide it contains the measurement, and where it is narrow it generally does not. Both halves of that sentence matter. At $Re = 100,000$ an analyst who reports agreement between XFOIL and experiment has reported their own choice of Ncrit, because a choice that agrees was always available. At $Re = 500,000$ no choice of Ncrit reaches the measurement on the E387 or the SD2030 — but the residual disagreement there (about 13 and 8 points) sits inside the window-choice spread of Table 3, so it is not a finding either.

The practical boundary falls near $Re = 200,000$ to $350,000$. Below it, a free-transition XFOIL result is not a prediction unless Ncrit is independently justified. Above it, the default is harmless.

5.5 Transition locations and the surface decomposition

XFOIL reports where it placed transition on each surface. Reading those out turns the previous section's result into a mechanism.

airfoil	Re	upper x/c	lower x/c	trip moves upper by
E387	100,000	0.567	1.000	0.547c
E387	200,000	0.490	1.000	0.470c
E387	350,000	0.385	1.000	0.365c
E387	500,000	0.356	1.000	0.336c
FX 63-137	100,000	0.692	0.859	0.672c
FX 63-137	200,000	0.578	0.803	0.558c
FX 63-137	350,000	0.511	0.742	0.491c

airfoil	Re	upper x/c	lower x/c	trip moves upper by
FX 63-137	500,000	0.455	0.711	0.435c
SD2030	100,000	0.624	1.000	0.604c
SD2030	200,000	0.551	1.000	0.531c
SD2030	350,000	0.492	0.998	0.472c
SD2030	500,000	0.451	0.991	0.431c

Table 7. XFOIL clean transition locations at $N_{crit} = 9$, averaged over $\alpha = 0-8^\circ$. A lower-surface value of 1.000 means no transition anywhere: laminar to the trailing edge.

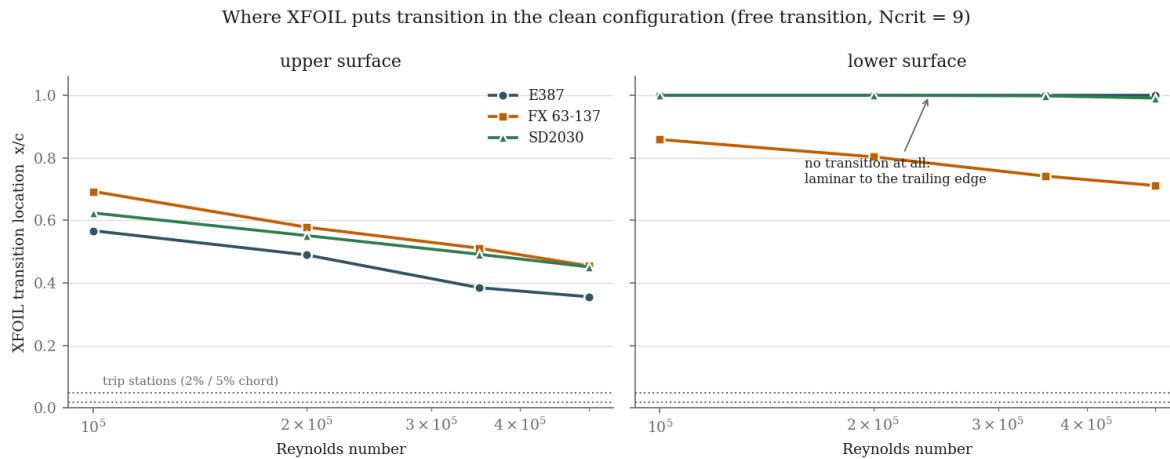


Figure 5. The same locations plotted. On the lower surface the E387 and SD2030 traces coincide at 1.000 and are indistinguishable.

On the E387 and SD2030 the lower surface never transitions at any Reynolds number tested. The trip at 5% chord therefore converts a fully laminar lower surface into a fully turbulent one, which is the largest change available. Splitting the trip by surface tests whether that is where the predicted penalty comes from.

airfoil	both surfaces	upper only	lower only	measured
E387	+9.9%	-0.9%	+10.6%	-11.4%
FX 63-137	+4.6%	+0.5%	+2.0%	-25.9%
SD2030	+12.4%	+0.1%	+12.5%	-3.0%

Table 8. Predicted trip effect at $Re = 100,000$ and $N_{crit} = 9$, decomposed by surface. Upper-only forces transition at 2% chord and leaves the lower surface free; lower-only the reverse.

Tripping the upper surface moves transition forward by more than half a chord and changes drag by less than one percent on all three airfoils. The entire predicted penalty is the lower surface. The upper-surface trip costs nothing because two effects cancel: it eliminates the laminar separation bubble, which saves drag, and it adds turbulent skin friction over the extra 55% of chord, which costs about the same. On the lower surface, where XFOIL predicts no bubble at all, there is nothing to save and the skin friction is paid in full.

That cancellation is delicate, and N_{crit} controls it. A larger bubble means the trip saves more while costing the same:

N_{crit}	clean Cd	upper-tripped Cd	upper-only effect
7	0.01857	0.02020	+9.9%

Ncrit	clean Cd	upper-tripped Cd	upper-only effect
9	0.02041	0.02020	-0.9%
11	0.02388	0.02020	-15.6%
13	0.03296	0.02020	-36.0%

Table 9. E387 at $Re = 100,000$: the upper-surface trip alone, against Ncrit. The tripped value is constant because with the upper surface forced and the lower surface never transitioning, no part of the solution depends on the amplification factor.

At $N_{crit} = 13$ tripping the upper surface saves 36% of the drag. That is the same sign and the same order of magnitude as the measured net effect of -11.4%, and it is the physics the wind tunnel is reporting: at $Re = 100,000$ the E387 carries a bubble large enough that destroying it is worth more than the turbulent friction that replaces it.

5.6 Calibrating Ncrit, and what that reveals

Sections 5.4 and 5.5 treat Ncrit as an unknown and bound its effect. The complementary question is what value the data itself prefers. For each airfoil at each Reynolds number, the clean-drag error is a monotone function of Ncrit, so there is generally a value that makes it zero. Fitting it is a test of whether Ncrit means anything physical: it represents the tunnel's turbulence level, so one value should work for every airfoil at every Reynolds number in that tunnel.

airfoil	Re	measured Cd	best Ncrit	fit found?	error at N = 9	note
E387	100k	0.02488	11.2	yes	-15.8%	bubble drag under-resolved
E387	200k	0.01343	5.5	yes	+1.3%	—
E387	300k	0.01199	—	no	+6.5%	over-predicts at every Ncrit
E387	350k	0.01087	—	no	+7.9%	over-predicts at every Ncrit
E387	500k	0.00969	—	no	+10.5%	over-predicts at every Ncrit
FX 63-137	100k	0.05070	12.7	yes	-50.0%	
FX 63-137	150k	0.02197	12.2	yes	-14.3%	
FX 63-137	200k	0.01750	12.0	yes	-7.8%	
FX 63-137	350k	0.01349	7.4	yes	-2.3%	
FX 63-137	500k	0.01219	5.6	yes	+1.1%	
SD2030	100k	0.01816	7.1	yes	+6.2%	
SD2030	200k	0.01319	8.3	yes	+1.8%	
SD2030	350k	0.01054	6.8	yes	-0.6%	
SD2030	500k	0.00960	7.2	yes	-1.6%	

Table 10. Ncrit fitted to the measured clean drag, 0–8° window, searched over Ncrit = 5 to 14. “Fit found” means the error changed sign inside that range; where it did not, no admissible Ncrit reproduces the measurement.

It does not work, and it fails in three different ways.

- SD2030 fits 7.1, 8.3, 6.8 and 7.2 at the four Reynolds numbers — essentially one value, 7.4 ± 0.7 . This is what a physical parameter looks like.
- FX 63-137 fits 12.7, 12.2, 12.0, 7.4 and 5.6, sliding monotonically downward as Reynolds number rises. Same tunnel, same model, same air. A number that has to fall from 12.7 to 5.6 to keep a model honest is not describing the tunnel; it is absorbing a Reynolds-dependent error in the model.
- E387 fits 11.2 at Re = 100,000 and 5.5 at Re = 200,000 — the difference between a sailplane and a dirty wind tunnel, one Reynolds number apart — and above Re = 300,000 there is no fit at all: XFOIL over-predicts the drag at every Ncrit from 5 to 14. The knob runs out of range.

The three airfoils also disagree with each other. If Ncrit were the turbulence level of the UIUC tunnel, all three would fit the same number. SD2030 says 7.4, the FX says anything from 5.6 to 12.7, the E387 says 5.5 to 11.2 or nothing. They cannot all be right about the same tunnel.

And the fitted values sit outside what the physics permits. Mack's correlation on $Tu < 0.1\%$ requires N_{crit} above 8.1; SD2030's consistent best fit is 7.4 and the FX at high Reynolds number wants 5.6. Fitting N_{crit} to drag data produces values the tunnel's own flow quality forbids, which is the clearest possible sign that the parameter is absorbing something that is not turbulence.

6. Lift, and why it matters more than drag

Everything above is drag. Drag is what a wake rake measures well and what a validation study naturally reaches for, but it is not the quantity a wing is bought for. Section 7 shows that a given percentage error in maximum lift costs more than five times what the same percentage error in drag costs on an autocross lap, so the drag focus of sections 2 through 5 was aimed at the less important variable.

Lift data is available for the SD2030 from the same Volume 4 campaign, and for the S1223 from Volume 2. The S1223 is included deliberately: it is the high-lift airfoil Formula SAE teams actually build wings from, so it is the most directly useful comparison in this document. Coordinates for both are in the AeroSandbox database, repanelled as before. Maximum lift is taken as the peak of the upstroke before the first collapse, by the same rule used to reject post-stall XFOIL solutions.

airfoil	Re	exp CLmax	at	XFOIL	at	CLmax error	lift slope error
S1223	149,500	2.205	17.1°	2.291	13.0°	+3.9%	+31.1%
S1223	198,900	2.209	16.2°	2.289	12.0°	+3.6%	+28.2%
SD2030	100,000	1.074	11.3°	1.176	10.0°	+9.5%	+14.1%
SD2030	200,000	1.148	12.3°	1.248	11.0°	+8.7%	+4.7%
SD2030	350,000	1.145	10.4°	1.297	12.0°	+13.3%	+2.4%
SD2030	500,000	1.189	11.4°	1.333	12.0°	+12.1%	+2.3%

Table 11. Lift comparison at $N_{crit} = 9$. Lift slope is fitted over $\alpha = -2^\circ$ to 6° . The SD2030 $Re = 350,076$ block is excluded: it is one of the short blocks noted in section 2.5 that went through a different reduction.

Three results, and all three are consistent in sign across both airfoils and every Reynolds number.

- XFOIL over-predicts maximum lift by 3.6% to 13.3%, always. For a wing designer this is the dangerous direction: the tool promises more downforce than the wing delivers.
- XFOIL's lift-curve slope is too steep by 2.3% to 31%, worst at low Reynolds number and worst on the high-lift S1223. Thin-airfoil theory gives 0.110 per degree; XFOIL returns 0.116 for the S1223 while the tunnel measures 0.0885, about 19% below ideal. A real high-lift airfoil at these Reynolds numbers is carrying a bubble and some trailing-edge separation through its whole working range, and XFOIL is not.
- On the S1223, XFOIL stalls 4.1° to 4.2° early — 13° against a measured 17° . That error is in the safe direction, but it is four degrees of usable angle of attack that a team designing to XFOIL would never know it had.

Selig's own caveat about sidewall boundary-layer separation affecting high-lift airfoils applies to the S1223 more than to anything else in this study, so the lift-slope discrepancy should not be read as entirely XFOIL's.

7. Findings

Finding 1 — the reliable comparison is the tripped one, and it is good above $Re \approx 350,000$

With transition forced, XFOIL has no free parameter and predicts drag to within about 5% at $Re = 350,000$ and above on all three airfoils. At $Re = 100,000$ it is off by between 4% and 23%, with inconsistent sign. This is the only claim in the study that cannot be altered by a parameter someone else chooses differently, and it is the one to design against.

Finding 2 — with transition free, the low-Reynolds-number prediction is set by N_{crit} , not by the airfoil

Sweeping N_{crit} from 7 to 13 at $Re = 100,000$ moves the predicted trip effect by 23 to 67 percentage points and reverses its sign on all three airfoils, while the measurement of course does not move. Every airfoil crosses its measured value inside the range that $Tu < 0.1\%$ permits, so at that Reynolds number an XFOIL result that matches the tunnel is reporting the analyst's N_{crit} , not the airfoil's behaviour.

The effect is confined to low Reynolds number, which revision 2 asserted and revision 3 measures. The N_{crit} -reachable range narrows monotonically — 23 to 67 points at $Re = 100,000$, 17 to 30 at $Re = 150,000$ – $200,000$, 3 to 6 at $Re = 500,000$ — and stops containing the measurement above $Re \approx 200,000$ on two of the three airfoils. The practical boundary is between $Re = 200,000$ and $350,000$.

The first revision of this document reported that XFOIL over-states the drag cost of tripping at $Re = 100,000$. That claim is true at $N_{crit} = 9$ and false at $N_{crit} = 12$, and nothing in the source documentation chooses between them. It is withdrawn as a statement about XFOIL and retained only as a statement about XFOIL at its default setting.

The practical form: below $Re \approx 200,000$, a free-transition XFOIL result is not a prediction unless it is accompanied by an independently justified N_{crit} . Above $Re \approx 350,000$ the sensitivity is small enough that the default is harmless.

Finding 3 — the mechanism, which the first revision could not identify

The predicted trip penalty at $Re = 100,000$ comes almost entirely from the lower surface, where XFOIL at $N_{crit} = 9$ predicts laminar flow to the trailing edge and the trip therefore buys nothing while paying full turbulent skin friction. The upper-surface trip is nearly free because bubble elimination and added friction cancel. Raising N_{crit} enlarges the bubble until the upper-surface trip becomes a large net saving — -36% at $N_{crit} = 13$ — which has the sign and roughly the magnitude of the measured net effect.

So XFOIL is not wrong about the mechanism. It is wrong about how much drag the laminar separation bubble carries at its default amplification factor, and it is very likely also wrong to predict a lower surface that never transitions at all at $Re = 100,000$ under an adverse pressure gradient. Those are specific, physical, testable criticisms. The first revision could offer only “the error is on the clean side for one airfoil and the tripped side for another, and we do not know why.”

Finding 4 — two methodological results, both of which changed answers here

The angle-of-attack averaging window moves results by more than the measurement uncertainty does at every Reynolds number above $200,000$, and by enough at $Re = 100,000$ to erase one claimed finding entirely: the SD2030 measured trip effect runs from -9.6% to $+0.4\%$ depending on the window, so it has no sign. Any single number from a study like this is meaningless without the window stated alongside it.

And a solver's convergence must be checked rather than assumed. Three points in 182 failed silently, and one of them alone reversed a headline number. The signature is specific and worth recognising: a quantity whose value depends on how many iterations were allowed is not a result. The same failure appears in root-finding, in time-stepping, and in every iterative method — it is not an XFOIL quirk.

A residual check is necessary and not sufficient. The FX 63-137 point at $Re = 500,000$ and $\alpha = 6^\circ$ converged cleanly to a tripped drag half the size of the same quantity at every other N_{crit} , in a case where N_{crit} cannot legitimately matter at all. It was caught by an invariant, not by a tolerance. Where a calculation has a quantity that must not change, computing it anyway and checking that it does not is worth more than any amount of residual monitoring — and it costs nothing here, because the control run had to be done regardless.

Finding 5 — lift is predicted worse than drag, in the dangerous direction, and matters more

XFOIL over-predicts maximum lift by 3.6% to 13.3% on both airfoils tested and at every Reynolds number, and over-predicts the lift-curve slope by up to 31%. Section 8 converts those into lap time and finds that a percentage error in lift costs more than five times what the same error in drag costs. A validation effort that measures only drag — which is what this study was for its first four revisions — is checking the wrong quantity.

8. What this means for a Formula SAE wing

A 250 mm chord element at 15 m/s sits at $Re \approx 250,000$, just below the Reynolds number where the tripped comparison becomes trustworthy. To turn the error magnitudes above into something a team can act on, they are propagated through a quasi-steady lap simulator — corner-speed limit from tyre load sensitivity, forward pass on traction and power against drag, backward pass on braking — on a 560 m autocross-style layout with a 320 kg car at $CL_A = 3.5$ and $CD_A = 1.3$. A 22 km endurance is 39 laps of it. The question asked is the designer's question: if XFOIL is wrong by the measured amount and the car is built believing it, what does that cost?

quantity wrong	by	lap (s)	lap penalty (s)	endurance penalty (s)
drag	+5%	27.471	+0.014	+0.6
drag	+12.5%	27.492	+0.035	+1.4
drag	+20%	27.514	+0.057	+2.2
drag	+25%	27.528	+0.072	+2.8
maximum lift	-3.6%	27.514	+0.058	+2.3
maximum lift	-8.7%	27.596	+0.139	+5.5
maximum lift	-12.1%	27.651	+0.194	+7.6
maximum lift	-13.3%	27.670	+0.213	+8.4
both, realistic	+5% / -9%	27.615	+0.158	+6.2

Table 12. Consequence of the measured XFOIL errors, propagated through the lap simulator. The last row combines a 5% drag under-prediction with a 9% maximum-lift over-prediction, which is the realistic case at FSAE Reynolds number.

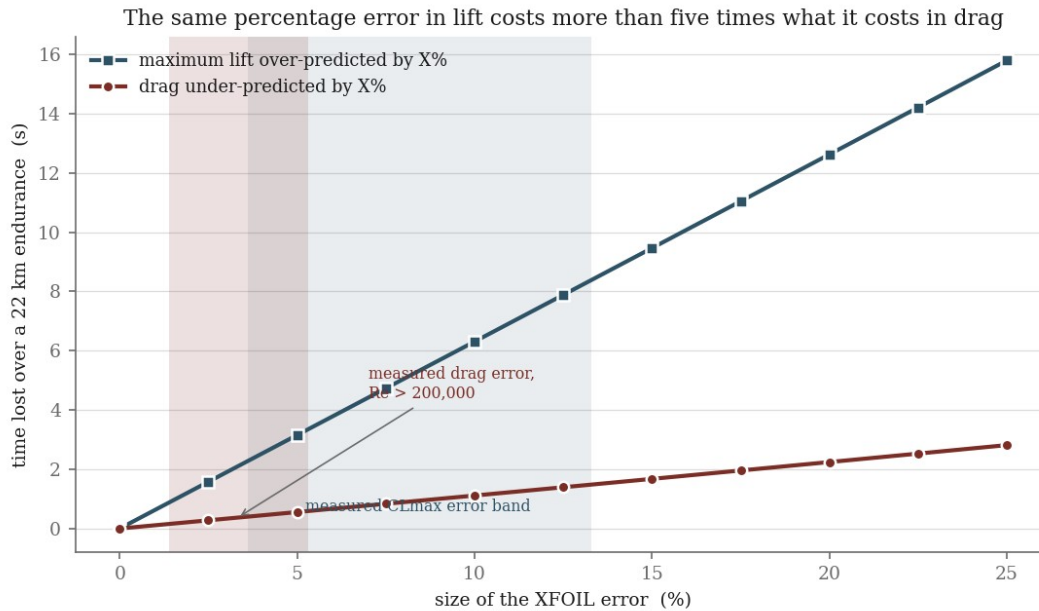


Figure 6. The same result plotted against error size, with the measured error bands shaded.

A 20% drag error costs 2.2 seconds over an endurance run. A 20% lift error costs 12.6 seconds. The ratio is 5.6, and the reason is that an autocross car is cornering-limited almost everywhere: downforce sets corner speed, and drag only spends money on the few short straights. Four revisions of this document have been about drag.

Four things follow for a team.

First, the realistic combined case — XFOIL under-predicting drag by about 5% and over-predicting maximum lift by about 9% at FSAE Reynolds number — costs about 0.16 s per lap and 6.2 s over endurance. That is not catastrophic, but it is systematic and it is in the optimistic direction: the car is slower than the model said, every lap, and the error never averages out.

Second, effort should go where the sensitivity is. Measuring or bounding maximum lift is worth roughly five times as much as the same work on drag. A team with limited testing time should spend it establishing what CL-A the car actually makes, not refining a drag polar.

Third, the Ncrit appropriate to a racing car is not 9. Nine describes an average wind tunnel. A car on a hot autocross surface behind other cars, with a wing in the wake of the front tyres, sits closer to Drela's "dirty tunnel" range of 4 to 8 — which Tables 5 and 6 say predicts a much larger penalty for any transition-forcing feature, and a shorter laminar run to protect in the first place. Running a polar at Ncrit = 9 and then racing in dirty air uses the wrong number in the flattering direction.

Fourth, the laminar run is worth protecting and is easy to destroy. Table 7 says roughly 45% of the chord is laminar at FSAE Reynolds numbers in XFOIL's estimate. A bug strike, a paint ridge, a badly placed sticker or a fingerprint at 2% chord converts that to turbulent flow, and Table 8 says what that costs — on the lower surface, a lot.

And one caution that section 2.6 forces. The lap-time numbers above are only as good as the aerodynamic inputs, and those inputs came from one wind tunnel whose E387 results differ from three other tunnels by up to 25% at $Re = 200,000$. The sensitivities in Table 12 are sound; the absolute penalties carry that scatter.

9. Threats to validity

- Three airfoils, one facility. Every measurement here comes from the same tunnel, the same wake rake, the same builder and the same reduction program. A systematic error in that chain would appear in this study as a systematic error in XFOIL. Selig's own comparison of E387 results against NASA Langley LTPT, Delft and Stuttgart is the available check on this, and it is not reproduced here.
- Selig notes separately that sidewall boundary-layer separation can affect results for high-lift airfoils, which bears on the FX 63-137 more than on the other two.
- The Ncrit sweep covers four values (7, 9, 11, 13) at four Reynolds numbers. It bounds the reachable range but does not locate the correct value, which would require a measurement of the tunnel's turbulence intensity rather than the upper bound the documentation gives.
- Two rows of Table 6 — FX 63-137 at $Re = 100,000$ and $500,000$ — survive on two angles of attack, because that airfoil is where XFOIL's convergence is worst. Both are marked, and neither carries any claim on its own.
- The XFOIL trip model forces transition at a station; it does not model a physical zigzag element with its own thickness and form drag. XFOIL should therefore under-state the trip penalty on that account alone, which makes the over-statement at $N_{crit} = 9$ slightly worse than it appears, and the agreement at $N_{crit} = 13$ slightly better than it deserves.
- Transition location is a computed quantity throughout section 5.5, never a measured one. The Volume 4 files record lift, drag and moment only. The claim that XFOIL's laminar lower surface is wrong is inference, not observation, and the measurement that would settle it — oil-flow visualisation, which UIUC published for some airfoils in Volumes 1 and 2 — has not been consulted.
- Coordinates come from the AeroSandbox database, not from the models as built.

10. Reproducing this

Data: Selig et al., Summary of Low-Speed Airfoil Data, Volume 4 (NREL/SR-500-34515), clean and tripped drag files for E387, FX 63-137 and SD2030.

Software: XFOIL 6.99 (available as a Debian package — `apt-get install xfoil`); Python 3.11 with numpy, matplotlib and aerosandbox.

Scripts, in the order they run:

<code>parse_v4.py</code>	read the UIUC .DRG format, variable spanwise column count
<code>trip2.py</code>	one XFOIL point per subprocess, with convergence screening
<code>rung4_raw.py</code>	sweep all airfoils / Reynolds numbers / angles, dump raw JSON
<code>tables.py</code>	window sensitivity, overlap test, Tables 1-3
<code>pconv_sd.py</code>	panel convergence check, Table 4
<code>ncrit_final.py</code>	Ncrit sweep at every Reynolds number, both screens, Table 6
<code>Rung4_Sessions_7_8.ipynb</code>	Ncrit and transition decomposition, Tables 5, 7-9

The raw JSON holds every accepted and rejected point, so the tables can be rebuilt without re-running XFOIL, and a different averaging window or overlap criterion can be applied to the same numbers. The notebook runs end to end in about 75 seconds.

11. Revision history

Revision 4 — the current document:

- Section 2.6 adds the four-facility E387 comparison from Volume 1, which quantifies the one-tunnel weakness instead of merely listing it. It supports the low-Reynolds-number findings and undercuts the few-percent agreements at high Reynolds number.
- Section 5.6 fits N_{crit} to the measured clean drag. One airfoil gives a stable value, one drifts monotonically with Reynolds number, one has no admissible value above $Re = 300,000$, and the three disagree with each other — so N_{crit} is not measuring the tunnel.
- Section 6 adds lift and maximum lift, the first non-drag validation in the study. XFOIL over-predicts CL_{max} by 3.6–13.3% on both airfoils at every Reynolds number, and stalls the S1223 four degrees early.
- Section 8 is rebuilt around a lap-time propagation. A percentage error in lift costs more than five times what the same error in drag costs, which means revisions 1 through 3 concentrated on the less important quantity. This is stated as a finding rather than quietly corrected.

Revision 3:

- The N_{crit} sweep is extended from $Re = 100,000$ to every Reynolds number (section 5.4, Table 6, Figure 4), closing the threat revision 2 listed against itself. The result supports what revision 2 assumed: the sensitivity collapses with Reynolds number, and the practical boundary is $Re \approx 200,000$ to 350,000.
- A second screening stage is added (section 3.4). The tripped configuration's independence from N_{crit} is an invariant, and checking it caught a solution that passed the residual test while being wrong by a factor of two. One angle of attack is rejected on this ground.
- Finding 4 gains the corresponding methodological point: a residual check is necessary and not sufficient.

Revision 2:

- Finding 1 of revision 1 — that XFOIL over-states the drag cost of tripping at $Re = 100,000$ — is withdrawn as a general claim and retained only as a claim about $N_{crit} = 9$. The N_{crit} sweep of section 5.4 is new.
- The tripped-configuration comparison, previously reported as a supporting table, is promoted to the study's primary result, on the grounds that it is the only comparison with no free parameter.
- Finding 3, previously an open question about differing mechanisms, is answered by the surface decomposition of section 5.5.
- Section 2.4 on stated flow quality, and section 3.2 on N_{crit} , are new.
- The FSAE implications in section 7 are rewritten. Revision 1 argued that XFOIL's low-Reynolds-number error was systematic and therefore correctable by calibration; that argument assumed the error was a property of the code rather than of an unmeasured input, and it does not survive section 5.4.

References

Selig, M. S., Guglielmo, J. J., Broeren, A. P., and Giguère, P., Summary of Low-Speed Airfoil Data, Volume 1, SoarTech Publications, Virginia Beach, 1995. Sections 2.1 and 2.4.

Selig, M. S., and McGranahan, B. D., Wind Tunnel Aerodynamic Tests of Six Airfoils for Use on Small Wind Turbines, NREL/SR-500-34515, National Renewable Energy Laboratory, 2004. Also AIAA 2004-1188.

Drela, M., XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils, Low Reynolds Number Aerodynamics, Springer, 1989.

Mack, L. M., Transition and Laminar Instability, NASA JPL Publication 77-15, 1977. Source of the N_{crit} -turbulence intensity correlation.

Coleman, H. W., and Steele, W. G., Experimentation and Uncertainty Analysis for Engineers, Wiley.